

Travaux de Dwork

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V
 $\downarrow$ smooth proper
 $\text{Spec } \mathbb{F}_p[t]$

$\lambda \in \mathbb{F}_{p^n}$, look at V_λ

$$Z(V_\lambda, t) = \exp \sum \frac{\# V_\lambda(\mathbb{F}_{p^n})}{n} t^n$$

$$= \prod_{i=0}^{2 \dim} \det(1 - Ft | H^i) (-1)^{i+1} (1 - t)$$

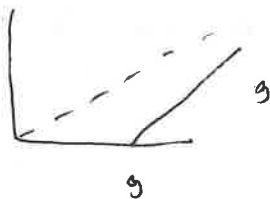
Idea: ~ how does Z vary p -adically?

p -adic power series: $a(x) = \sum a_n x^n$ s.t. $a_n \rightarrow 0$ p -adic norm

For each $\bar{\lambda} \in \mathbb{F}_{p^n}$, consider $\lambda \in W(\mathbb{F}_{p^n})$ Teichmüller lift,

then I want $a(\lambda) a(\lambda^p) a(\lambda^{p^2}) \dots a(\lambda^{p^{n-1}})$ to be some specified $1/\text{root}$ of the char. poly. of Frob.

Ex. If C is an ordinary curve,



C is ordinary if Newton polygon of $F \sim H^{\text{cris}}$
 = Hodge

$$Z = \frac{(1 - \alpha_1 t) \dots (1 - \alpha_g t) (1 - \beta_1 t) \dots (1 - \beta_g t)}{(1 - qt)(1 - t)}$$

$$\text{val}_p(\alpha_i) = 0, \quad \text{val}_p(\beta_j) = 1, \quad \beta_i = q/\alpha_i$$

k perfect field of char p , $W = W(k)$

$S = \text{Spec } A$ smooth affine W -scheme

$S^n = \text{Spec}(A/\varpi^n A)$ smooth over W_n

S^∞ formal completion $S^\infty = \text{Spf } A$ p -adic formal scheme.

$$\begin{array}{ccc}
 \phi \hookrightarrow S^\infty & \longleftarrow & S^0 \supset \sigma \\
 \text{many such } \phi \downarrow \tilde{\sigma} & \begin{array}{c} \downarrow \\ \downarrow \\ \downarrow \end{array} & \downarrow \uparrow e \\
 \text{Spf } W & \longleftarrow & \text{Spf } k \supset \sigma \\
 \cup & & \\
 \sigma & &
 \end{array}
 \quad \text{s.t.} \quad \tilde{\sigma} \circ \sigma = \phi \circ \tilde{e}$$

H locally free on S^∞ , and ∇ an integrable connection nilpotent (continuous)

$\nabla_{\tilde{\sigma}}$, operator on H , topologically nilpotent.

Def. An F -crystal (H, ∇, F) is a locally free H with ∇ nilpotent

as above, for every lift of Frob $\phi: S^\infty \rightarrow S^\infty$, there is a map

$F(\phi): \phi^* H \rightarrow H$ which is flat and becomes an isom. after inverting p .

require some compatibility between $F(\phi_1)$ and $F(\phi_2)$.



$k = \mathbb{F}_q$, then $\sigma^n = \text{id}$ on $W(\mathbb{F}_{p^n})$, if $S^\infty = W$, then F iterated n times is

an endomorphism of H , then $P(H) := \det(1 - F^n | H)$

Prop 3.1 Let (H, ∇, F) be an F -crystal over $W(k)[[t_1, \dots, t_n]]$.

(1) Let $W(k) \ll t_1, \dots, t_n \gg$ ring of convergent divided power series over $W(k)$

$H \otimes W(k) \ll t_1, \dots, t_n \gg$ has a flat basis of solutions.

(2) $K = \text{Frac } W(k)$, $K \langle\langle t_1, \dots, t_n \rangle\rangle$ ring of power series over K
convergent on the open unit disc

$$\sum a_{k_1, \dots, k_n} t_1^{k_1} \dots t_n^{k_n} \text{ s.t. } |a_{k_1, \dots, k_n}|_p r^{k_1 + \dots + k_n} \rightarrow 0$$

for any $0 < r < 1$.

(3) $H \otimes K \langle\langle t_1, \dots, t_n \rangle\rangle$ has a basis of flat sections.

Moreover if $\alpha \in H \otimes W(k) \ll t_1, \dots, t_n \gg$ s.t. $F(\psi)(\psi^* \alpha) = \alpha$, then α lies in H .

Rem. this shows "most" diff. ops do not come from an F -crystal.

$f(t) = \exp(t^p)$, $f^* = pt^{p-1}f$ violates (2) although it is nilpotent.

$\alpha \in \mathcal{O}_{\mathbb{A}^1_p}$, $f(\alpha)$ converges if $|\alpha|_p \leq 1/p^{p(p-1)}$

§ Hermitian F -crystals

$f: X \rightarrow S^u$ smooth proper geom. integral fibres,

Assume that dR coh is locally free

Th $H^i = R^i f_* \Omega_{X/S^u}^i$ equipped with ∇_{dR} , then this underlies an F -crystal

F is given by functoriality of crystalline cohomology.

For $\bar{\lambda} \in \mathbb{F}_q^n$, $\lambda \in W(\mathbb{F}_q^n)$ Teichmüller lift

generally $\lambda \in S^w$ a Teich lift: then $\zeta(X_\lambda, t) = \prod_{i=1}^{2 \dim} \det(1 - Ft | H^i)^{(-1)^{i+1}}$

Then Mazur proved: F-crystal H^i is divisible.

$F(\varphi) \varphi^* \text{Fil}^1 \subset p H^i$, then

$$H^i / (\text{Fil}^1 + p H^i) = R^i f_* \Omega_{X^0/S^0} / \text{Fil}^1 = R^i f_* \mathcal{O}_{X^0}.$$

This gives a semilinear map. $\varphi^* R^i f_* \mathcal{O}_{X^0} \rightarrow R^i f_* \mathcal{O}_{X^0}$

Dual to the Cartier operator.

In ordinary case, this map is an isom.

For Hodge F-crystal H and I assume that the length of the 0 slope part is

Then $\exists$ subcrystal $U \subset H$ s.t. U and Fil^n are transversal.

and s.t. $F(\varphi): \varphi^* U \rightarrow U$ is an isom (as modules $H \cong \text{Fil}^n \oplus U$ not with connection)
and slopes of $H/U \geq 1$ (not just after inverting p)

U is the piece of H of unit slope.



Let $s_1, \dots, s_r \in W(k) \langle t_1, \dots, t_r \rangle$ be flat sections spanning U .

$$F(\varphi) \varphi^* s_j = M_{ij} s_i, \quad M_{ij} \text{ is invertible.}$$

~~Ex Legendre family $y^2 = x(x-1)(x-\lambda)$~~

Suppose that λ is a $W(\mathbb{F}_{p^n})$ point of S^{ns} s.t. $\varphi^n = \text{id}$ over λ

char poly. of Frob on U_λ is $F(\varphi) \cdot \varphi^* F(\varphi) \cdots (\varphi^*)^{n-1} F(\varphi)$

$$: (\varphi^*)^n U \rightarrow U$$

char poly of this $(M(\lambda) \cdot M(\lambda^p) \cdots M(\lambda^{p^{n-1}}))$

Ex. Legendre family $y^2 = x(x-1)(x-\lambda)$

$$\begin{array}{c} E \\ \downarrow \end{array}$$

$$\text{Spec } \mathbb{Z}_p[\lambda, (\lambda(\lambda-1))^{-1}]$$

M is just a scalar $= 1/a(\lambda)$

$$T = \frac{1}{a(\lambda)} \cdot \frac{1}{a(\lambda^p)} \cdots \frac{1}{a(\lambda^{p^{n-1}})}$$

$\lambda \in W(\mathbb{F}_{p^n})$ Teich.

$$\text{let } (1-t) \quad t = a(\lambda) \cdots a(\lambda^{p^{n-1}})$$

$$Z(E_\lambda) = \frac{(1-t)(1-p^n/at)}{(1-t)(1-p^n t)}$$

$$\lambda(\lambda-1) \frac{d^2 f}{d\lambda^2} + (2\lambda-1) \frac{df}{d\lambda} + \frac{1}{4} f = 0$$

$$F(1/2, 1/2, 1, \lambda) = \sum_{j=0}^{\infty} \binom{-1/2}{j} \lambda^j$$

unique sol'n ^{hol.} near $\lambda=0$

$$\text{it turns out that } a(\lambda) = \frac{F(1/2, 1/2, 1, \lambda)}{F(1/2, 1/2, 1, \lambda^p)}$$

then $a(\lambda)$ p -adic converges
at radius ≥ 1

and is the a on LHS.

$$\# E_{\bar{\lambda}}(\mathbb{F}_p) = a(\lambda) + p/a(\lambda), \quad \bar{\lambda} \in \mathbb{F}_p$$

$$\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{1}{2} m \frac{d}{dt} (v^2)$$

the mass m is constant, so $\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{1}{2} m \frac{d}{dt} (v^2)$

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